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BAGONG PILIPINAS

CC 101 – Introduction to Computing

Laboratory Activity No. 1 | Week 2: Introduction to Computing (CO1) | Unit 2, Computing Career Pathways

Explore Career Pathways

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Course/Program: BSCS 1-1 **Date:** September 16, 2026

I. Objectives

At the end of this activity, you should be able to:

- Identify the major computing career pathways introduced in lecture and describe what each one involves.
- Choose one computing career pathway that best matches your interests and explain why.
- Map a realistic entry-to-expert roadmap for your chosen pathway, including representative job titles and key skills at each stage.
- Compare your current skills against your chosen pathway's requirements and identify a concrete plan to close the gap.

II. Materials Needed

- Your CC 101 Unit 2 lecture slides/notes on Computing Career Pathways
- This worksheet (printed copy, or the digital file to fill out and submit)
- Optional: internet access, for Part E (real job posting search)

III. Background

Computing is not just "programming." As introduced in the Unit 2 lecture, the field branches into several distinct career pathways, each with its own operational scope, required technical competencies, representative job titles, and professional standard or reference body: Software Engineering, Systems & Cloud Infrastructure, Data Science & AI Engineering, Cybersecurity & Threat Defense, and Information Systems & Enterprise IT. Each pathway also has its own career roadmap — the entry-level, mid-level, senior-level, and expert/leadership roles a professional typically grows through. This activity asks you to review those five pathways, choose the one that interests you most, and map out what growing in that pathway could look like for you.

IV. Procedure

Part A – Recall the Five Pathways

1. Using your lecture notes or slides, complete Table 1 by writing the core operational scope and one to two representative job titles for each of the five computing career pathways.

Part B – Choose Your Pathway

2. From the five pathways, choose ONE that interests you the most.
3. In two to three sentences, explain why you chose it (e.g., skills you already enjoy, a subject you like, a job title that excites you).

Part C – Map Your Career Roadmap

4. If your chosen pathway is Software Engineering, Systems & Cloud Infrastructure, Data Science & AI Engineering, or Cybersecurity & Threat Defense, use the career roadmap shown for it in the lecture. If

you chose Information Systems & Enterprise IT (no roadmap was shown in lecture), research typical entry-level, mid-level, senior-level, and expert/leadership roles for this pathway online.

5. Complete Table 2 with example job titles and key skills/responsibilities at each of the four career stages.

Part D – Skill Gap Self-Assessment

6. Looking at the essential technical competencies for your chosen pathway (Table 1 / lecture), list at least five in Table 3. For each, mark whether you already have some experience with it, and briefly note how you plan to learn it if you don't.

Part E – Real-World Job Search (Optional / Bonus)

7. Search a real job platform (e.g., LinkedIn, Indeed, JobStreet, Kalibrr) for one entry-level job posting that matches your chosen pathway. Record it in Table 4.

V. Data and Findings

Table 1. The Five Computing Career Pathways (Recap — Part A)

Pathway	Core Operational Scope	Representative Job Title(s)
Software Engineering	Building applications, designing, testing and improve computer programs and apps that people use	Software Developer, Application Programmer
Systems & Cloud Infrastructure	Maintain computer and to maintain hardware networks, servers, and cloud storage and also to secure	Systems Administrator, Cloud Support Engineer
Data Science & AI Engineering	Build systems that learn and make predictions, extracting insights from structured/unstructured data and training predictive AI models	Data Scientist, Machine Learning Engineer
Cybersecurity & Threat Defense	Protecting computers, networks, and data from being hacked or stolen, safeguarding network	Cybersecurity Analyst, Information Security Analyst
Information Systems & Enterprise IT	Manage technology, aligning business workflows with technology solutions and enterprise software architectures	Business Systems Solution Architect Analyst, IT Manager,

My Chosen Pathway (Part B)

I am most interested in: Data Science & AI Engineering

Why I chose it:

I chose data science and ai engineering because i love finding answers hidden in information and looking for patterns, i enjoy organizing details and turning numbers into something meaningful. Being a data analyst sounds really exciting to me because i can help make smart decisions based on facts.

Table 2. My Career Roadmap for My Chosen Pathway (Part C)

Career Stage	Example Job Title(s)	Key Skills / Responsibilities
Entry-Level	Junior Data Analyst, Data Science Intern	Entry level use excel and python and clean and organize data, make simple charts and also they follow instructions from senior team members
Mid-Level	Data Analyst, AI Developer	Build dashboards and create basic ai models and explain results clearly to the team
Senior-Level	Senior Data Scientist, Lead AI Engineer Lead	Lead big projects and design advance system and guide mentor satff
Expert / Leadership	Chief Data Officer, AI Research Director	They make long term plans and manage teams, budgets also decide how the company uses data and ai

Table 3. Skill Gap Self-Assessment (Part D)

Essential Competency	Already Have It? (Y/N)	How I Plan to Learn It
Basic computer & internet skills	Yes	Keep practicing basic skills in computer, explore new tools regularly
Organizing & presenting information	Yes	Practice making reports and charts
Python / programming basics	Yes	Take free online lessons, practice basic programming
Statistics & math for data	Yes	Review basic math, start with simple stats tutorials
Critical thinking & problem-solving	Yes	Practice analyzing real examples of problem solving
Microsoft Excel / spreadsheets	Yes	Watch simple tutorials and practice every week

Table 4. Real Job Posting Found (Part E, Optional / Bonus)

Item	Details
Job Title	Junior Data Analyst

Item	Details
Company	DataHub Sulotions
Platform / Source (e.g., LinkedIn, Indeed, JobStreet)	JobStreet
Required Skills Listed (3–5)	1. Basic Excel 2. Good with numbers 3. Willing to learn 4. Attention to details 5. Basic computer skills

VI. Realization (Reflection)

In two to four sentences each, answer the following in your own words.

1. Why did you choose this particular pathway over the other four?

I choose data science & ai engineering because I really like working with facts and finding answers from them. The other pathways are good too, but this one feels more like what I enjoy doing. I also think it will help me get a good job in the future.

2. Looking at your roadmap, which career stage excites you the most, and why?

The career stage mid-level stage excites me most because i will be doing real work on my own. I won't just be learning anymore but actually create things that can help people. It feels like the point where i truly become good at what i do.

3. Which required skill or competency for this pathway do you already have, and which one will be hardest for you to develop?

The required skill or competency for this pathway i already have is i'm already good at organizing things and paying attention to small details. Learning programming will be the hardest for me because it's something totally new. But i'm ready to practice every day until i get it.

4. How does mapping this roadmap change — or confirm — your plans for the rest of this course and your degree?

It makes my goals much clearer and more real to me. Now I know exactly what skills I need to focus on starting right now. It motivates me to do well in this course because i can see where it leads me.

VII. Scoring Guide

Criteria	Description	Points
Completeness	Tables 1–3 and the reflection are fully filled out (Table 4 is bonus).	25
Roadmap Accuracy & Depth	Roadmap and skills recorded match the lecture pathway (or credible outside research for IS & Enterprise IT).	25
Realization / Reflection	Answers are thoughtful, specific to the student's own choice, and well written.	30
Neatness & Timeliness	Worksheet is legible/well-formatted and submitted on time.	20
Total		**100**

Checked by: **Score:** / 100

Instructor's Signature over Printed Name / Date